Speech Emotion Recognition using Deep Learning

Jyothisha B
gmail: jyothishab445@gmail.com

***Abstract*— Speech Emotion Recognition (SER) is a fast-emerging field that seeks to recognize and interpret human emotions using vocal signals. Speech nuances rhythm, tone, and pitch contain dense emotional content capable of affecting perception and behaviour. This study delves into SER as a critical element in enriching technologies such as intelligent virtual assistants, mental health diagnosis, and affective computing systems. Using a series of benchmark datasets such as RAVDESS and TESS, the study preprocesses and normalizes the audio input, converting it to log-Mel spectrogram representations for significant feature extraction. State-of-the-art deep learning structures, specifically Convolutional Neural Networks (CNNs), in collaboration with traditional and optimized feature extraction methods, are utilized to extract emotional content from speech. Experimental findings verify the practical feasibility of SER systems in real applications, demonstrating their capability to capture and classify emotional states from vocal content effectively.**

Keywords*— SER, CNN, MFCC, Speech Emotion Recognition, Deep Learning, Audio Signal Processing, Feature Extraction, RAVDESS, TESS

I. Introduction

Speech is the most intuitive and natural human communication. Of all the modalities, voice signals are most readily accepted as among the fastest and most efficient human-computer interactions. While human beings can readily identify emotional information in speech through various sensory inputs, it is a challenging task for machines. Identification of emotions in speech with high accuracy enhances the quality and richness of human-machine communication. Language-based Speech Emotion Recognition (SER) is a very critical task to differentiate emotional patterns in various speaker characteristics, such as gender differences. Features like Linear Predictive Cepstral Coefficients (LPCC), Mel-Frequency Cepstral Coefficients (MFCC), and fundamental frequency components are typical analyzed features in SER and serve as the foundation for emotion perception studies.

Yet, isolating emotions from speech is inevitably made challenging by differences in speaking styles, rates of speech, sentence forms, and speaker identities. The very same spoken statement can produce varied emotional responses in each listener, further complicating its recognition. Environmental and cultural considerations also play a significant role in the articulations of emotions in speech, adding another layer of variability. Emotional states may be generally categorized as short-term and long-term emotions, yet most SER systems are not precise enough to indicate these variations. Recognition systems can either be speaker-dependent or speaker-independent, and each of them has its own challenges in emotional classification. This study delves into some of the machine learning and deep learning methods in SER, i.e., K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). The research article begins with an introduction to speech emotion recognition and proceeds with a system architecture explanation. It goes on with the discussion of most widely used emotional speech databases such as RAVDESS and TESS, emotional speech analysis techniques, and feature extraction techniques such as log-Mel spectrograms. Under methodology, CNN-based models and optimal feature extraction methods are demonstrated to emphasize SER's potential in real-world applications. This study takes inspiration from the potential of SER in enhancing human-computer interaction, cognitive analysis, and mental disorder diagnosis. The research article concludes with a thorough analysis of results and an explanation of areas for future development.

II. Literature Review

Speech Emotion Recognition (SER) has been a research topic of ongoing interest in the last few decades, with several advances having been reported in modeling, feature extraction, and classification. Several seminal studies have investigated the subtleties of emotion recognition from speech through both conventional machine learning and deep learning methods.Aharon et al. [1] examined para-linguistic characteristics using a deep neural network that was convolutional and recurrent. Their trained model on speech representations in spectrograms achieved a 68% recognition rate on the IEMOCAP dataset with a high-complexity ConvLSTM model. This implies the significance of integrating temporal and spatial information in SER tasks.Jonathan et al. [2] introduced a multitask learning method combined with deep convolutional generative adversarial networks (DCGANs) to produce large amounts of unlabeled data. Their method expanded the training corpus size to 100 hours, thereby improving emotion classification accuracy by 43.88%, illustrating the potential of big data in model generalization.Chen et al. [3] introduced delta and delta-delta features and Mel spectrograms to extract more emotional cues and remove irrelevant features. Their 3D attention-based ACRNN obtained high unweighted average recall rates on IEMOCAP and Emo-DB, selectively focusing on emotionally significant frames.Zhao et al. [4] proposed a 1D-2D branch hybrid CNN model to learn from raw audio and log-Mel spectrograms. With Bayesian hyperparameter optimization and transfer learning for speeding up training, their deep CNN fusion model achieved improved accuracy on benchmark datasets.Yenigalla et al. [5] emphasized the affective importance of spectrograms and phoneme sequences compared to text input. Their hybrid CNN using both phoneme and spectrogram features produced good performance on the IEMOCAP dataset and outperforms existing approaches in overall and average class accuracy.Sarma et al. [6] tested a variety of DNN structures for emotion recognition from time-domain and frequency-domain features, i.e., high-dimensional MFCCs. Their work showed that learning time-domain features was better than frequency-based methods. Most importantly, their best-performing structure—a time-limited self-attention model—achieved 70.6% weighted accuracy, proving the importance of temporal aggregation.Latif et al. [7] explored cross-corpus and cross-language transfer learning using Deep Belief Networks (DBNs). On five three-language datasets, DBNs were superior to SVMs and sparse autoencoders in all of the experiments. Multi-lingual training on small target domain data was demonstrated to significantly improve SER performance.Zhao et al. [8] proposed 1D and 2D CNN+LSTM architectures to learn local and global emotional features from an audio. Their architecture with local feature learning blocks (LFLBs) and max pooling layers learned hierarchical and long-range dependencies efficiently and enhanced performance on benchmark datasets.Sun et al. [9] proposed a sparse autoencoder with attention mechanisms to distinguish emotional speech frames from neutral frames. Their method, applied to cross-language online speech databases, achieved better emotion prediction performance by concentrating on emotionally dense parts.Jiang et al. [10] employed deep learning to extract and fuse features from various acoustic groups. Their system had a fusion network for learning discriminative representations and followed that with the use of an SVM classifier. On the IEMOCAP dataset, their method gave a 64% boost over existing models.Pandey et al. [11] contrasted deep learning architectures for SER from Mel spectrograms, magnitude spectrograms, and MFCCs. The best emotional feature coupling performance was obtained, their analysis suggested, by hybrid models that combined CNNs and LSTMs.Zhen et al. [12] introduced a CNN-BLSTM-SVM pipeline using log-Mel spectrograms that outperformed current models for the IEMOCAP dataset. Although very effective, the generalizability of the model requires more experiments with larger datasets.A comparative study by [13] compared various SER architectures on six datasets and concluded that CNN+LSTM worked best on five out of six, demonstrating its robustness on various speech corpora.Guo et al. [14] proposed a Kernel Extreme Learning Machines (KELM) model based on spectral features. Although their approach performed well on Emo-DB and IEMOCAP, its specificity to the dataset limited its application in all settings. Misbah et al. [15] employed deep convolutional networks (DCNNs) for extracting log-Mel features from raw speech of corpora such as IEMOCAP, SAVEE, Emo-DB, and RAVDESS. They experimented with many classifiers—such as SVM, Random Forest, k-NN, and neural networks—and found that although a few performed well on one database, none were generalizable on all four.

III. METHODOLOGY

The main purpose of SER is to increase human-machine interaction. To create a model for speech emotion recognition, two datasets are used.

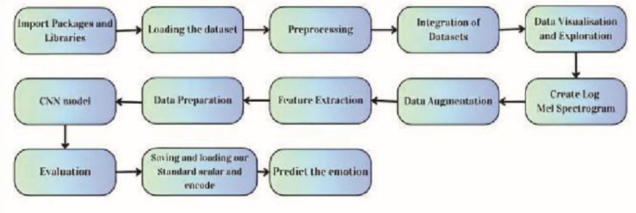

Fig. 1. Block Diagram

Working Model:

To begin with, it is crucial to remember that there are numerous platforms available for running and executing Python code. Several platforms are available, including

1. Google Colab
2. Kaggle
3. Pycharm
4. Jupyter Notebook

This study mostly focuses on Kaggle. Kaggle Notebooks is a cloud-based computing tool designed to support collaborative and repeatable analysis. The user has access to cutting-edge data analysis and machine learning programs that are pre-installed and compatible.

A. Loading the dataset

In this research on speech emotion recognition, the methodology is characterized by a systematic approach that starts with data imported from a RAVDESS and TESS dataset.

1) RAVDESS Dataset

The RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) includes 1,440 speech files recorded by 24 professional actors (12 male, 12 female), each vocalizing two neutral North American English statements with eight emotions (neutral, calm, happy, sad, angry, fearful, disgust, and surprised) at two intensity levels (normal and strong). Each file follows a 7-part naming convention encoding metadata such as modality, emotion, intensity, statement, repetition, and actor ID. This structured format facilitates efficient dataset organization and preprocessing for Speech Emotion Recognition (SER) tasks.

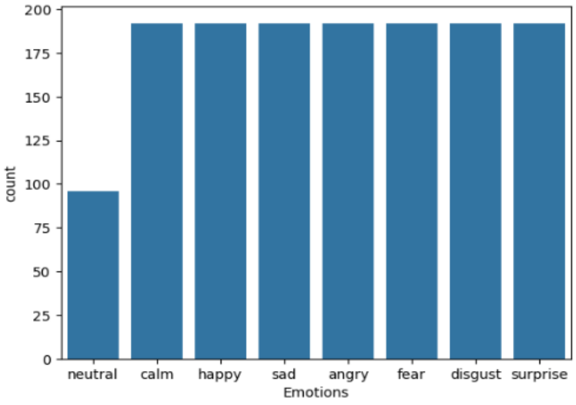

Fig. 2. RAVDESS Dataset distribution

2)TESS dataset

TESS stands for Toronto emotional speech set [16] in which two actresses (ages 26 and 64) spoke a set of 200 target words with the phrase "Say the word _." These words were recorded as they showed each of seven emotions: anger, disgust, fear, happiness, happy surprise, sadness, and neutral. There are a total of 2800 data points, which are audio files. The information is set up so that each of the two female actors and their feelings are in their folders. That's where you can find the audio file with all 200 target words. The audio file is saved in the WAV format.

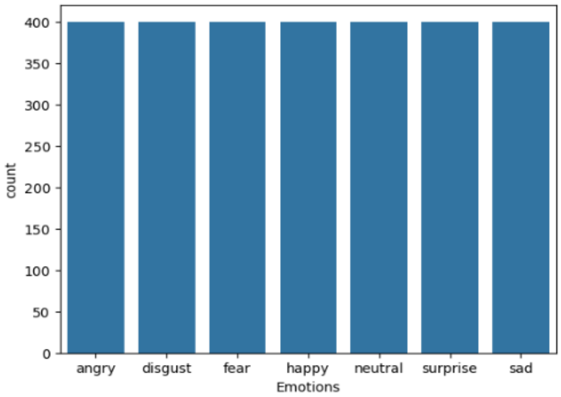

Fig.3.TESS dataset distribution

3. Merging RAVDESS and TESS Datasets

This research merges the RAVDESS and TESS datasets for Speech Emotion Recognition (SER). The RAVDESS dataset contains 1,440 audio-only speech files from 24 actors (12 male, 12 female) speaking eight emotions neutral, calm, happy, sad, angry, fearful, disgust, and surprised at two levels of intensity. Files have a structured 7-part naming convention to encode metadata for efficient preprocessing. The TESS dataset comprises 2,800 female speakers' audio recordings (ages 26 and 64) each pronouncing 200 target words within the context of "Say the word _," spoken under seven emotions: anger, disgust, fear, happiness, happy surprise, sadness, and neutral.To maintain uniformity and enhance model performance, the 'calm' emotion label was eliminated from both datasets to achieve uniformity among emotional categories for model training.

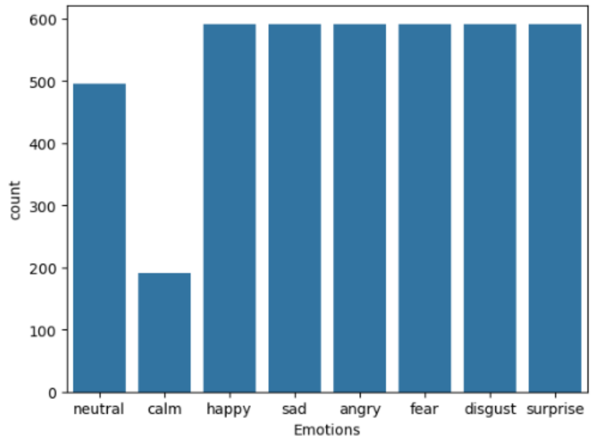

Fig.4. Merged dataset distribution

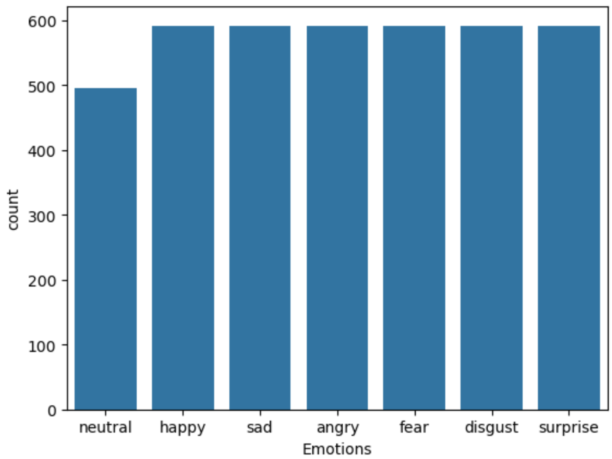

Fig.5. After removal of calm label

B. Preprocessing

The imported audio files are then pre-processed from the Ravdess,,TESS dataset. Integration ensures consistency across datasets.

C. Integration of Dataset

TABLE I. INSTANCES AND EMOTIONS

Name: Emotions, dtype: int64

disgust	592
fear	592
sad	592
happy	592
angry	592
neutral	496
surprise	592

Table I indicates that, for instance, there are 592 instances labelled as "Disgust," 592 instances labelled as "Fear," and so on.

D. Data augmentation

Data augmentation techniques, including noise addition, stretching, shifting, and pitch alterations, are implemented to bolster the model's resilience.

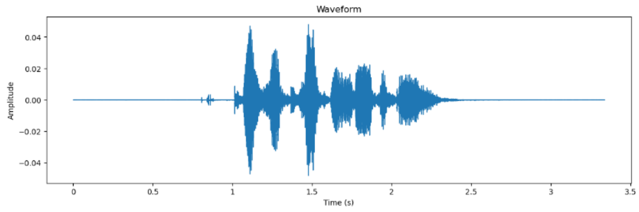

Fig. 6. Normal Audio

Fig 6 offers both a visual representation of the audio waveform and the ability to play the audio directly within the Jupyter Notebook or IPython environment.

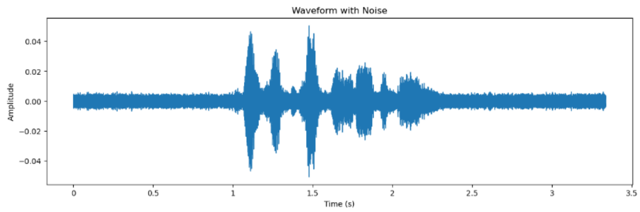

Fig. 7. Audio with Noise

Fig 7 provides both a visual representation of the noisy audio waveform and the ability to listen to the audio with added noise.

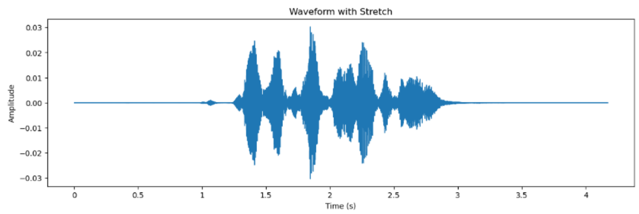

Fig. 8. Stretched Audio

Fig 8 provides both a visual representation of the stretched audio waveform and the ability to listen to the stretched audio.

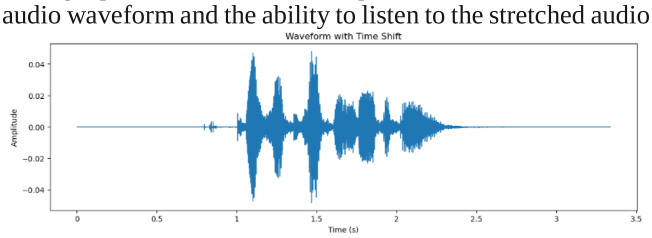

Fig.9.Time Shifted Audio

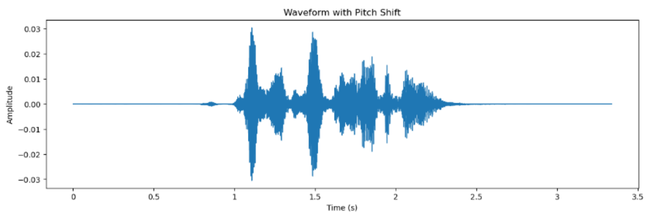

Fig.10.Pitch Shifted Audio

Fig 9 ,10 provides both a visual representation of the shifted audio waveform and the ability to listen to the shifted audio.

Shifting the audio in time introduces a time delay, which can be observed in the waveform plot and heard when playing the audio.

E. Feature Extraction

Feature extraction is performed on the pre-processed audio data using both standard and optimized methods to capture pertinent information for emotion recognition. Conventional and optimized approaches to feature extraction are explored, providing a baseline for comparison and improving computational efficiency.

Extracted features are saved to facilitate future experiments, and data preparation involves separating features and labels, applying one-hot encoding for multiclass classification, reshaping for compatibility with CNN models, and scaling using Sklearn's Standard Scaler. Early stopping is employed across all models during training.

The feature techniques that we are used:

1. Zero Crossing Rate (ZCR)
   What it measures: The rate at which the audio signal passes over the zero amplitude line.
   Why it is important: It is applied to differentiate unvoiced and voiced phonemes. High ZCR is used to denote percussion or noise phonemes, while low ZCR denotes speech or harmonic phonemes.

2. Chroma STFT (Chroma Short-Time Fourier Transform)
   What it does: Captures the spread of musical notes (pitches) in a sound signal, irrespective of octaves.
   Why it matters: It is useful when breaking down music because it recognizes harmonies and chords.

3. MFCC (Mel-Frequency Cepstral Coefficients)
   What it does: Imitates the vocal tract shape to mimic human auditory perception.
   Why it matters: It's the gold standard for speech and speaker recognition because it outperforms others at capturing timbral features.

4. RMS Energy (Root Mean Square Energy)
   What it does: Compares the overall loudness of a signal.
   Why it matters: Helps to identify silent, soft, and loud areas of an audio file.

5. Mel Spectrogram
   What it does: Translates the energy of different frequency components across time onto the Mel scale.
   Why it matters: It is a compact representation of audio material and is used extensively in deep learning networks for sound classification.

	0	1	2	3	4	5	6
0	2.269350	-0.026034	-0.637117	-1.215732	-1.052938	-1.342330	-1.339570
1	-0.802824	0.456877	0.652721	0.767439	0.579603	0.773996	1.069725
2	-0.536397	0.435009	0.510279	0.767439	0.734031	0.359504	0.175769
3	1.661891	-0.695836	-0.916780	-1.164935	-1.163244	-1.088607	-0.988175
4	-0.587963	0.431575	0.592046	0.609276	0.565439	0.872030	1.051035

Fig. 11. Features

```
        150       151      152       153      154  labels
0 -0.977492 -0.955861 -0.91281 -0.807886 -0.68662       0
1 -0.977492 -0.955861 -0.91281 -0.807886 -0.68662       1
2 -0.977492 -0.955861 -0.91281 -0.807886 -0.68662       0
3 -0.977492 -0.955861 -0.91281 -0.807886 -0.68662       4
4 -0.977492 -0.955861 -0.91281 -0.807886 -0.68662       2
```

Fig. 12. Features and Emotions

Fig 11 and 12 show the output with columns representing different features, and the 'lables' column containing the corresponding emotion labels. The head of the data frame is displayed, providing a glimpse of the data structure.

The research encompasses the implementation of a CNN model for speech emotion recognition, leveraging the preprocessed and scaled data. A Convolutional Neural Network (CNN) model is also developed, predictions are made on the test data, and random predictions are scrutinized for validation. Various plots are generated to visualize the performance of multi-models.

F. Predict the Emotions

TABLE II. PREDICTION TABLE

	Predicted Labels	**Actual Labels**
1.	angry	angry
2.	disgust	disgust
3.	angry	angry
4.	neutral	neutral
5.	fear	Fear
6.	angry	angry
7.	Fear	fear
8.	angry	angry
9.	sad	sad
10.	sad	sad

Table II displays the predicted labels and actual labels for the initial 10 samples in the test set. It offers a concise summary of the model's performance on the test data. It is making predictions on the test data using a trained model. Itthen compares the predicted labels with the actual labels and creates a data frame to display the results.

G. Evaluation

Evaluation ensues, rigorously assessing the performance of different models, and results from the best-performing model are scrutinized based on predefined metrics. The best model is serialized to JSON and HDF5 formats, with the weights of the model also serialized for future use. Additionally, the Standard Scaler object used for scaling is saved, ensuring consistency in preprocessing. This methodology concludes with the development of a test script enabling the model to predict new records, showcasing its practical application in real-world scenarios.

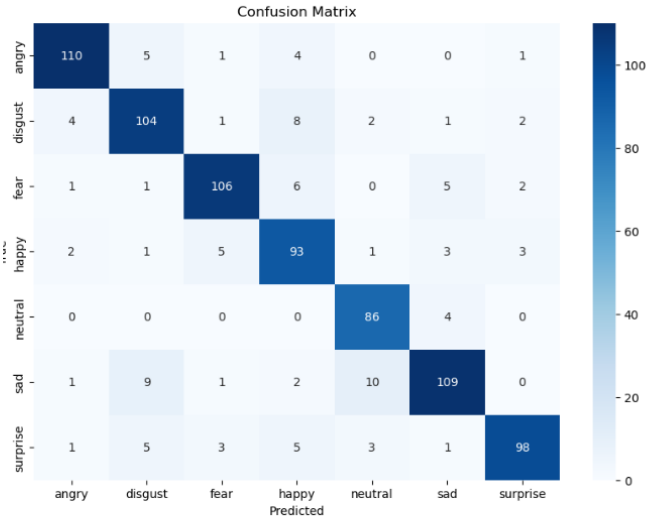

Fig. 13. Confusion Matrix

Figure 13 displays the result of the code, presenting a visual representation of the confusion matrix and a textual summary of the classification report. The confusion matrix heatmap provides a visual representation of the model's performance in accurately and inaccurately identifying cases for each emotion class. The categorization report offers further metrics to assess the model's performance on individual emotion classes.

TABLE III. PREDICTED LABELS

	Precision	Recall	F1-score	support
Angry	0.92	0.91	0.92	121
Disgust	0.83	0.85	0.84	122
Fear	0.91	0.88	0.89	121
Happy	0.79	0.86	0.82	108
Neutral	0.84	0.96	0.90	90
Sad	0.89	0.83	0.85	132
Surprise	0.92	0.84	0.88	116
Accuracy			0.87	810
Macro avg	0.87	0.87	0.87	810
Weighted avg	0.87	0.87	0.87	810

Table III shows the classification report provides additional metrics for evaluating the model's performance on each emotion class.

IV. RESULT ANALYSIS

Speech emotion recognition analysis involves evaluating the performance of your machine learning model. This process includes dividing your data into training, validation, and testing sets, training the model, and assessing it using key metrics like accuracy, precision, recall, and F1-score. Visualizations such as confusion matrices provide insights into the model's performance. Error analysis helps identify patterns in misclassifications. Comparative analysis against benchmark models offers context. Optionally, examine bias and fairness. Qualitative analysis, listening to audio samples, confirms real-world applicability. Iterate to improve the model's accuracy and generalization, ensuring it aligns with human judgment in recognizing emotions in speech.

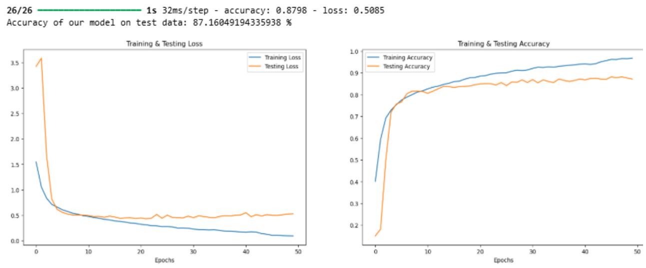

Fig. 14. Accuracy

Figure 14 displays the model's accuracy on the test data. Visual representations of the trends in training and testing loss and accuracy across epochs are the outputs. You can examine the model's performance in learning from the training data and in generalizing to the unseen test data from these charts.

V. CONCLUSION

In this paper, our exploration has encompassed an array of models and methodologies, contributing to the collective understanding of SER systems and their potential applications.

The study combines and preprocesses audio data using a variety of datasets, such as RAVDESS, TESS, uses log mel spectrograms to effectively extract features. Methods like CNN models are used, along with standard and optimized feature extraction techniques. The results show how good the suggested methods are at effectively distinguishing between speech and emotional signals, and they also show how important SER is in realworld applications. This research positions itself within the dynamic landscape of SER, contributing not only to the refinement of existing models but also to the identification of critical challenges that demand collective attention. To propel SER systems to new heights, concerted efforts are required to address these challenges. Future research endeavours could focus on the development of more diverse and inclusive datasets, encompassing a broader spectrum of languages, accents, and emotional expressions. Additionally, exploring novel feature extraction techniques, including the incorporation of contextual information and multimodal cues, may enhance the discriminative power of SER models. Interdisciplinary collaborations, involving linguists, psychologists, and computer scientists, can contribute valuable insights for a more comprehensive understanding of speech emotion dynamics.

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